ECN 7055 Macroéconomie B

Luca Mazzone

Cours II.4: Dynamique d'Entreprise

Université de Montréal

Introduction to Firm Dynamics

Dynamic Quantitative Models of Firm Behavior:

- Focus on firms as key players in macroeconomic models
- Thinks of role of firms in investment, production, employment, and economic growth
- Focus on measurable ingredients of firm level decisions such as technology and costs

Two Empirical Benchmarks

1. Zipf 's law
   - frequency of observation inversely proportional to rank (word-use, city-size, firm-size)

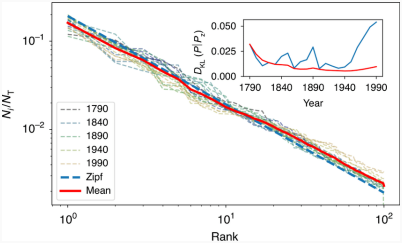

Figure 1: City Size Distribution

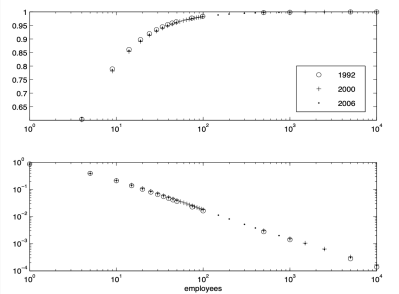

Figure 2: Firm Size Distribution

2. Gibrat's law
   - individual firm / city growth rates independent of size (at least, for large enough firms / cities)

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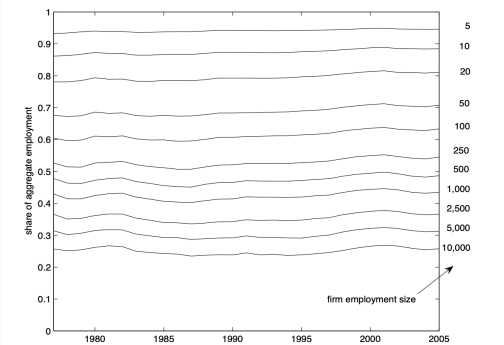

Figure 1: Shares of Aggregate Employment

Firm size as a Pareto distribution

- Suppose we measure firm size by number of employees, n.
- Firm-size distribution is well approximated by a Pareto distribution.

A Reminder on Pareto Distributions:

- Survivor function for standard Pareto distribution (lower bound normalized to 1):

$$\Pr[n_0 \geq n] = n^{-\xi}, \qquad n > 1,\ \xi > 0.$$

- Associated CDF and PDF:

$$F(n) = 1 - n^{-\xi}, \qquad f(n) = \xi n^{-(\xi+1)}.$$

- Finite mean requires $\xi > 1$, finite variance requires $\xi > 2$, finite skewness requires $\xi > 3$, etc.
- Zeta distribution is the discrete analogue. Zipf's law is the zeta distribution with $\xi = 1$.

Facts: stable and skewed distribution

- In short, US (same in other countries!) firm-size distribution is both very stable and very skewed:
  - there are about 6 million firms
  - around one-half of total employment is accounted for by some 18,000 very large firms with more than 500 employees each
  - around one-quarter of total employment is accounted for by some 1,000 enormous firms with more than 10,000 employees each
  - but most firms are small: almost 80% of firms have less than 10 employees
- Perhaps surprisingly, a stable firm-size distribution is the fairly natural consequence of random growth at the micro level (almost).

Gibrat's law

- Starting point: Gibrat's law of 'proportional effect'.
- Fixed population of units i (cities, firms, ...) of size P_t^i.
- Let $S_t^i := P_t^i / \bar{P}_t$ denote normalized size (relative to average size $\bar{P}_t$) and suppose

$$S_{t+1}^i = \varepsilon_t^i S_t^i, \qquad \varepsilon_t^i \text{ i.i.d. with density } f(\varepsilon).$$

- Growth independent of current size $\Rightarrow$ absolute increment $S_{t+1} - S_t$ approximately proportional to current size S_t.
- Let $G_t(x) := \Pr[S_t^i \geq x]$ denote the survivor function.

From random growth to Pareto distribution

- Law of motion for the survivor function:

$$G_{t+1}(x) = \Pr[\varepsilon S_t \geq x] = \Pr\left[S_t \geq \frac{x}{\varepsilon}\right] = \int_0^\infty G_t\left(\frac{x}{\varepsilon}\right) f(\varepsilon)\, d\varepsilon.$$

- A steady-state distribution, if it exists, satisfies

$$G(x) = \int_0^\infty G\left(\frac{x}{\varepsilon}\right) f(\varepsilon)\, d\varepsilon.$$

- Guess-and-verify that $G(x) = kx^{-\xi}$ solves this fixed point problem.
- This requires:

$$1 = \int_0^\infty \varepsilon^\xi f(\varepsilon)\, d\varepsilon,$$

  which pins down the Pareto exponent ξ in terms of $f(\cdot)$. Coefficient k pinned down by requiring the density integrates to 1.

Existence of a steady-state distribution

- Of course there need not be a steady-state distribution.
- Suppose $f(\varepsilon)$ is log-normal, $\ln \varepsilon \sim N(\mu, \sigma^2)$, then

$$\ln S_t - \ln S_0 \sim N(\mu t, \sigma^2 t),$$

  and there is no steady-state distribution; it keeps fanning out.
- But 'small departures' from this strict version of Gibrat's law give us back a steady-state distribution and moreover give micro foundations for $\xi \approx 1$.
- Examples: Gabaix (1999 *QJE*), Luttmer (2007 *QJE*).

Lucas (1978) span of control

- Firm consists of a production technology and a managerial technology.
- Production technology is standard concave CRS:

$$y = F(k, n) = f(k/n)\, n.$$

- Managerial technology: a manager of talent x produces

$$x\, g(y)$$

  units of output, where $g(\cdot)$ is strictly increasing and strictly concave.
- Manager is a fixed input. DRS of $g(\cdot)$ reflects limited span of control: best manager can't control everything; efficient for some resources to be controlled by next-best manager, etc.

Lucas (1978) span of control: profit maximization

- Profits for a firm with manager x facing factor prices w, r:

$$\pi(x) = \max_{k,n} \left\{ x\, g(F(k,n)) - rk - wn \right\}.$$

- Write $\ell := k/n$ (capital-labor ratio), so $F(k,n) = f(\ell)\, n$ and

$$\pi(x) = \max_{\ell,n} \left\{ x\, g(f(\ell)\, n) - r\ell n - wn \right\}.$$

- First order conditions (with $y = f(\ell)\, n$):

$$xg'(y)f'(\ell) = r, \qquad xg'(y)f(\ell) = r\ell + w.$$

- All firms choose the same $\ell = k/n$ independent of x; it solves:

$$\frac{f(\ell) - f'(\ell)\ell}{f'(\ell)} = \frac{w}{r}.$$

- Scale $y(x)$ then determined from $xg'(y)f'(\ell) = r$, which can be used to recover $n(x)$.

Implications of Gibrat's law

- Firm growth induced by changes in factor prices:

$$\frac{d}{dt} \ln n(x; w(t), r(t)) = \frac{n_w(x, w, r)}{n(x, w, r)} \dot{w}(t) + \frac{n_r(x, w, r)}{n(x, w, r)} \dot{r}(t).$$

- Strong form of Gibrat's law: this derivative is invariant to firm size (managerial talent). That is,

$$\frac{\partial}{\partial x} \left[\frac{n_w(x, w, r)}{n(x, w, r)} \dot{w}(t) + \frac{n_r(x, w, r)}{n(x, w, r)} \dot{r}(t) \right] = 0.$$

- For this to hold for all patterns of changing factor prices, must have both

$$\frac{\partial}{\partial x} \left(\frac{n_w(x, w, r)}{n(x, w, r)} \right) = 0, \qquad \frac{\partial}{\partial x} \left(\frac{n_r(x, w, r)}{n(x, w, r)} \right) = 0.$$

Implications of Gibrat's law (restriction on g)

- The condition
$$\frac{\partial}{\partial x}\left(\frac{n_w(x,w,r)}{n(x,w,r)}\right) = 0$$
is implicitly a restriction on the functional form of $g(\cdot)$.
- Calculating $n_w(x,w,r)$ and solving the resulting differential equation for $g(\cdot)$, Lucas finds that
$$g(y) = Ay^{\alpha}, \qquad A > 0, \quad 0 < \alpha < 1,$$
is the unique functional form consistent with the strong version of Gibrat's law.

Lucas (1978) example

- Suppose production function $y = n$, managerial technology xy^{α}, and managerial talent has Pareto CDF $1 - x^{-\xi}$.
- First order condition:

$$x\,\alpha\,y^{\alpha-1} = w \quad \Rightarrow \quad y(x) = n(x) = \left(\frac{\alpha x}{w}\right)^{\frac{1}{1-\alpha}}.$$

- Hence if managerial talent has Pareto distribution with exponent ξ, then firm size is also Pareto with exponent $\xi(1-\alpha)$.

Managerial selection

- Suppose an individual of talent x can opt for wage w or manage and earn income $\pi(x)$.
- Indifference condition:

$$\pi(x) = w.$$

- Cutoff managerial type $x^\star$ such that only $x > x^\star$ actively manage.
- Interpreting w as the opportunity cost of the manager (a fixed cost), the cutoff compares managerial revenue with fixed + variable costs of operating.

Firm Dynamics

Firm Choices & Constraints

- Key firm decisions:
  - Production y
  - Hiring and Firing l
  - Investment i
  - Dividends payout d
  - Debt b, equity issuance n
  - Other choices: Mergers, acquisitions, IPOs, R& D...
- Constraints:
  - Technology
  - Capital accumulation
  - Financial constraints
- We will focus on models where only a subset of these choices are available.

Technology & Production Function

- Firms produces one good (y) using capital k and labor l
- Production function:

$$y = zF(k, l)$$

- z captures firm (total factor) productivity, managerial ability (Lucas (1978)), or even "organizational capital"

Constant Return to Scale

- Constant returns to scale (CRS) implies that the exact scale of the firm is indeterminate since per-period profits are linear in the scale of production
- Firms can be of any size and produce any amount either
- With CRS technologies we simply use one single firm to represent the whole economy.
  - This is very useful for macro models when we are not interested in explaining cross-section facts.

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  - This is very useful for macro models when we are not interested in explaining cross-section facts.
- Constant Returns to Scale does not work well in models with heterogeneous firms! Why?
  1. Most productive firm(s) would take over the entire economy!
  2. Outcome: monopoly or oligopoly of one or a few firms, contradiction w/ perfect-competition

Decreasing Return to Scale

- To study differences across firms we generally need the profit function to exhibit decreasing returns to scale (DRS).
- Now the scale of production is well determinate since producing more, or less, than the optimal scale lowers profits
- Note: DRS in operating profits can follow from simply DRS technology or from from combining CRS production with monopoly power.
- With DRS we can have an economy with:
  - a fixed exogenous number of firms (perhaps just one) or,
  - an endogenous number determined by free entry and exit.

Introduction to Heterogeneous Firms

- Why study firm heterogeneity in macroeconomic models?
- Differences in:
  - Productivity and size (today)
  - Stock returns
  - Financing patterns
  - Competition, markups, innovation
- Importance of cross-sectional distributions in aggregate dynamics
  - Default by a few firms can feed back to the rest of the economy and affect aggregate values
  - Generally the composition of firm investment, or demand for financing can affect aggregate productivity, savings, and asset prices.

Theoretical Framework for Firm Growth

Consider firms with DRS that rent capital every period to maximize per period profits:

- **Jorgensonian Model without Adjustment Costs**:

$$k(z; R) = \left(\frac{\gamma z}{r + \delta} \right)^{\frac{1}{1-\gamma}}$$

- Firm growth rate:

$$\frac{d \ln k(z; r)}{dt} = \frac{g}{1 - \gamma}$$

- **Gibrat's Law**: Firm growth is independent of size (Lucas 1978)
- **Introducing adjustment costs** to explain deviations from Gibrat's Law
  - Could be convex adjustment costs, training costs,...

Firm Size Distribution

- **Axtell (2001)**: Empirical evidence on firm size distribution
- Power law (Zipf's Law) for firm employment sizes:

$$P(l > L) = \left(\frac{l}{L}\right)^{-\eta}$$

- Empirical estimates of **Pareto tail parameter** $\eta = 1.06$
- Matching firm size distributions in theoretical models

Firm Heterogeneity

A Simple Model

- Let's write a model where factors are CRS-aggregated, but production is DRS:

$$y_i = a_i F(\mathbf{x}_i)^\gamma$$

The Dual Problem

1. Solve the cost-minimizing combination of inputs for one unit of $F(\mathbf{x})$:

$$\mathbf{x}^* = \arg\min_{\mathbf{x}} \mathbf{p} \cdot \mathbf{x}\,, \text{ subject to: } F(\mathbf{x}) = 1 \implies \text{minimized unit cost: } c \equiv \mathbf{p} \cdot \mathbf{x}^*$$

2. Decide on the optimal scale of production. Define $m_i = F(\mathbf{x}_i)$ the choice of inputs:

$$\max_{m_i} a_i m_i^\gamma - c m_i \implies a_i m_i^{\gamma-1} = \frac{c}{\gamma} \implies y_i = \frac{c}{\gamma} m_i$$

A Simple Model/2

- Adding up for all i:

$$Y = \int y_i \, di = \frac{c}{\gamma} M \, ; M = \int m_i \, di = \int F(\mathbf{x}_i) \, di = F(\mathbf{X})$$

- Define:

$$A \equiv \left(\int a_i^{\frac{1}{1-\gamma}} \, di \right)^{1-\gamma} = \frac{c}{\gamma} \left(\int m_i \, di \right)^{1-\gamma} \Longrightarrow A = \frac{c}{\gamma} M^{1-\gamma} \Longrightarrow Y = AF(\mathbf{X})^{\gamma}$$

because $Y = \frac{c}{\gamma} M$

A Simple Model/3

$$Y = \left(\int a_i^{\frac{1}{1-\gamma}} \, di \right)^{1-\gamma} F(X)^{\gamma}$$

- This relationship can be viewed as the aggregate production function
- But: ***aggregate outcome is influenced by the distribution of*** a_i
- Suppose $\log(a_i) \sim N(\nu - \frac{\sigma^2}{2}, \sigma^2)$, with $\exp(\nu) = \int a_i \, di$. Then:

$$A = \exp\left(\nu + \frac{\gamma}{1-\gamma} \cdot \frac{1}{2}\sigma^2 \right)$$

$\Longrightarrow$ dispersion influences the A even when avg productivity is constant! Why?

  - a productive firm uses more input so it has a greater role in Y
  - When $\sigma \uparrow$, the economy has more room to allocate resources to more productive firms .

- Allocation of resources is key!

Reallocation

Reallocation

- Firm heterogeneity affects aggregates through the endogenous re-allocation of inputs
- When firm productivity changes over time, the reallocation of inputs affects A.
- Foster, Haltiwanger, and Krizan (2001): quantify impact of reallocation:
  - define productivity of establishment i in t as $a_{i,t}$
  - output-weighted average productivity $\bar{A}_t$ is: $\bar{A}_t \equiv \sum_i s_{i,t} a_{i,t}$, where $s_{i,t} = \frac{y_{i,t}}{\sum_i y_{i,t}}$

$$\bar{A}_t - \bar{A}_{t-1} = \Delta\bar{A}_t = \sum_{i \in C} s_{i,t-1} \Delta a_{i,t} + \sum_{i \in C} (a_{i,t} - \bar{A}_{t-1}) \Delta s_{i,t} + \sum_{i \in C} \Delta a_{i,t} \, \Delta s_{i,t} +$$
$$+ \sum_{i \in N} s_{i,t} (a_{i,t} - \bar{A}_{t-1}) - \sum_{i \in \chi} s_{i,t} (a_{i,t} - \bar{A}_{t-1})$$

- where C are continuing firms, N are entrants, χ are exiters

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$$+ \underbrace{\sum_{i\in N} s_{i,t}(a_{i,t} - \bar{A}_{t-1})}_{\text{contribution of entrants}} - \underbrace{\sum_{i\in \chi} s_{i,t}(a_{i,t} - \bar{A}_{t-1})}_{\text{contribution of exiting } i\text{'s}}$$

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- Foster et al. (2001) estimates:
  - 45% of the aggregate change in multifactor productivity is accounted by within-firm growth

Modeling the Cross Section

Models of Cross Section of Firms

- Equilibrium investment
  - Lucas and Prescott (1971)
- Dynamic Models of Industry Equilibrium - no investment
  - Jovanovic (1982) - Bayesian learning plus entry and exit
  - Hopenhayn (1992) , Hopenhayn and Rogerson (1993)- Competitive industry equilibrium entry and exit
  - Ericson and Pakes (1995,1998) - Adds non-competitive behavior and suitable for empirical estimation
- Quantitative Dynamic Models
  - Gomes (2001), Adds investment and financing frictions
  - Luttmer (2007) Balanced growth

Entry & Exit of Firms

- The easiest way to model entry: assume an initial entry cost, e to start a new firm
- **Modeling firm entry**:

$$E[v(z,k)] \geq e$$

- If our model forces firms to pay a fixed cost of production in every period it is possible to also get exit
- **Exit decisions**:

$$v(z,k) < 0$$

- Exit more likely for **small & low-productivity** firms
- Implications for **selection bias in observed firms**

Hopenhayn (1992): environment, timing, static problem

- Time $t = 0, 1, 2, \ldots$; continuum of firms; no strategic interactions.
- DRS $\Rightarrow$ well-defined firm size; idiosyncratic productivity z_t follows Markov kernel $f(z'|z)$.
- Prices (p, w) taken as given; choose numeraire $w = 1$; industry demand $D(p)$ given.

Technology and static profits (given p)

$$y = zF(n), \qquad \pi(z; p) \equiv \max_{n \geq 0} \Big\{ p\, zF(n) - n - k \Big\},$$

with fixed operating cost $k > 0$. Let $n(z; p)$, $y(z; p)$ be the argmax policies.

Timing within a period

- Incumbents choose stay/exit; potential entrants choose enter/not.
- Stay $\Rightarrow$ pay fixed cost k; enter $\Rightarrow$ pay entry cost k_e and draw initial $z \sim g(z)$.
- After paying, firms draw next-period productivity and then operate.

Industry supply (endogenous): $Y(p) = \int y(z; p)\, \mu(z)\, dz$; market clears: $Y(p) = D(p)$.

Incumbent dynamic program and exit cutoff (discrete time)

Let $v(z;p)$ be the value of an incumbent with current productivity z at price p and discount factor $\beta \in (0,1)$.

Bellman equation

$$v(z;p) = \pi(z;p) + \beta \max\left\{0, \int v(z';p)\, f(z'|z)\, dz'\right\}.$$

Continuation value and exit rule

$$\mathcal{C}(z;p) \equiv \int v(z';p)\, f(z'|z)\, dz'.$$

The optimal policy is a cutoff rule: there exists $z^\star(p)$ such that

$$\text{exit iff } \mathcal{C}(z;p) < 0 \quad \Longleftrightarrow \quad z < z^\star(p).$$

Interior cutoff condition

$$\mathcal{C}(z^\star(p);p) = \int v(z';p)\, f(z'|z^\star)\, dz' = 0.$$

Entry, free entry, and stationary equilibrium

Entrants draw $z \sim g(z)$ after paying $k_e > 0$ and start producing next period. Let $m \geq 0$ be the mass of entrants.

Free entry condition (discrete time)

$$\beta \int v(z;p)\, g(z)\, dz \;\leq\; k_e, \qquad \text{with equality if } m > 0.$$

Stationary equilibrium (objects and conditions) A stationary equilibrium is $(v(\cdot), y(\cdot), z^\star, \mu(\cdot), m, p)$ such that:

(i) Given p, $v(\cdot)$ solves the incumbent Bellman equation, and $y(\cdot)$ is the associated static policy; $z^\star = z^\star(p)$ is the exit cutoff.
(ii) Free entry holds: $\beta \int v(z;p) g(z)\, dz \leq k_e$ with equality if $m > 0$.
(iii) Goods market clears: $\int y(z;p)\mu(z)\, dz = D(p)$.
(iv) The cross-sectional distribution is stationary (law of motion with exit + entry; next slide).

Stationarity, discretization, and computation

Stationary distribution with exit and entry

$$\mu(z') = \int \Gamma(z'|z)\,\mu(z)\,dz + m\,g(z'), \qquad \Gamma(z'|z) = f(z'|z)\cdot \mathbf{1}\{z \geq z^\star(p)\}.$$

Discretize to n productivity nodes $\{z_i\}_{i=1}^n$

$$\boldsymbol{\mu} = \boldsymbol{\Gamma}(p)\boldsymbol{\mu} + m\boldsymbol{g} \quad \Rightarrow \quad \boldsymbol{\mu} = m(\boldsymbol{I} - \boldsymbol{\Gamma}(p))^{-1}\boldsymbol{g}.$$

(trick: m is linear!)

Computation (two nested scalars)

- Step A (price from free entry). Guess p. Solve incumbent DP $\Rightarrow v(\cdot; p)$ and cutoff $z^\star(p)$. Update p until $\beta \int v(z; p) g(z)\,dz = k_e$ (bisection works since LHS rises in p).
- Step B (mass of entrants from market clearing). Given $p^\star$, compute $\boldsymbol{\mu}(1, p^\star) = (\boldsymbol{I} - \boldsymbol{\Gamma}(p^\star))^{-1}\boldsymbol{g}$ and set

$$m^\star = \frac{D(p^\star)}{\sum_{i=1}^n y_i(p^\star)\,\mu_i(1, p^\star)}.$$

Static special case for intuition + comparative statics (selection vs price)

Static "once-and-for-all" productivity draw: $z \sim g(z)$ and $F(n) = n^\alpha$, $0 < \alpha < 1$, $w = 1$.

$$\pi(z;p) = \max_{n \geq 0}\{pzn^\alpha - n - k\}. \implies \text{FOC: } \alpha pzn^{\alpha-1} = 1, \text{ then:}$$

$$n(z;p) = (\alpha pz)^{\frac{1}{1-\alpha}}, \qquad \pi(z;p) = (1-\alpha)\alpha^{\frac{\alpha}{1-\alpha}}(pz)^{\frac{1}{1-\alpha}} - k.$$

Exit cutoff: $\pi(z^\star;p) = 0 \Rightarrow$ firms operate iff $z \geq z^\star$.

Free entry + eliminate p using the exit condition

$$(1-\beta)\frac{k_e}{k} = \beta \int_{z^\star}^{\infty} \left[\left(\frac{z}{z^\star}\right)^{\frac{1}{1-\alpha}} - 1\right] g(z)\, dz.$$

Comparative statics

- $k_e \uparrow \Rightarrow p^\star \uparrow$, $m^\star \downarrow$, and $z^\star \downarrow$ (weaker selection; more low-z firms survive).
- Firm-size distribution: ambiguous because
  - *price effect* ($p \uparrow$ raises $y(z;p), n(z;p)$ for incumbents),
  - *selection effect* ($z^\star \downarrow$ increases mass of low-z firms).

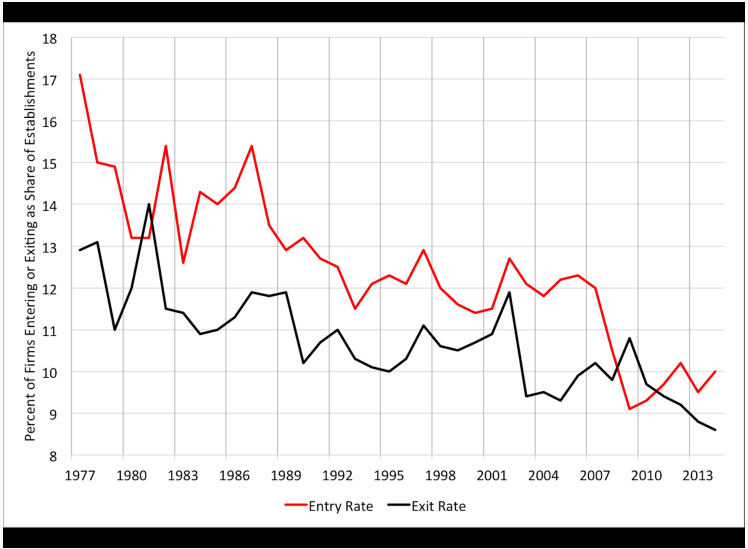
Firm-Level Data Sources

- Common datasets for firm heterogeneity studies:
  - **CRSP & Compustat**: Financial data
  - **SDC & Sageworks**: Additional panels
  - **U.S. Census Bureau**: Employment, productivity, industry-level data
  - **Compustat Global, Amadeus, Osiris**: International firm data
- Europe: many administrative datasets (Portugal, France, Italy, Sweden...)
- Importance of plant-level and establishment-level data

Empirical Evidence on Firm Size, Growth, and Survival

- **Evans (1988)** Firm size and growth relationships: smaller firms grow faster
- **Luttmer (2007)** Firm age and survival rates: small and young firms also exit faster
- Survivor function estimates for firms in:
  - U.S. manufacturing (Dunne, Roberts, Samuelson 1988)
  - Portuguese manufacturing (Mata & Portugal 1994)
  - U.S. employer firms (Headd 2003)
- Implications of firm heterogeneity for business dynamics

Establishment entry and exit rates from BDS



Misallocation

A Simple Model of Misallocation

- Back to the simple model. Government taxes firm i's output at rate τ_i. New problem:

$$\max_{m_i \geq 0} (1-\tau_i)\, a_i\, m_i^{\gamma} - c\, m_i \implies \text{FOC:} \quad (1-\tau_i)\, a_i\, \gamma\, m_i^{\gamma-1} = c$$

$$\Rightarrow \quad m_i = \left(\frac{\gamma}{c}\right)^{\frac{1}{1-\gamma}} a_i^{\frac{1}{1-\gamma}} (1-\tau_i)^{\frac{1}{1-\gamma}} \implies \quad y_i \equiv a_i m_i^{\gamma} = \left(\frac{\gamma}{c}\right)^{\frac{\gamma}{1-\gamma}} a_i^{\frac{1}{1-\gamma}} (1-\tau_i)^{\frac{\gamma}{1-\gamma}}.$$

- Aggregate "effective input" is still $M \equiv \int m_i\, di = \int F(\mathbf{x}_i)\, di = F(\mathbf{X})$, hence

$$M = \left(\frac{\gamma}{c}\right)^{\frac{1}{1-\gamma}} \underbrace{\int a_i^{\frac{1}{1-\gamma}} (1-\tau_i)^{\frac{1}{1-\gamma}}\, di}_{\equiv S_0} \quad \Rightarrow \quad \left(\frac{\gamma}{c}\right)^{\frac{\gamma}{1-\gamma}} = \left(\frac{M}{S_0}\right)^{\gamma}.$$

$$Y = \int y_i\, di = \left(\frac{\gamma}{c}\right)^{\frac{\gamma}{1-\gamma}} \underbrace{\int a_i^{\frac{1}{1-\gamma}} (1-\tau_i)^{\frac{\gamma}{1-\gamma}}\, di}_{\equiv S_1} = \left(\frac{S_1}{S_0^{\gamma}}\right) M^{\gamma} \implies \qquad Y = A\, F(\mathbf{X})^{\gamma},$$

$$A \equiv \frac{\int a_i^{\frac{1}{1-\gamma}} (1-\tau_i)^{\frac{\gamma}{1-\gamma}}\, di}{\left(\int a_i^{\frac{1}{1-\gamma}} (1-\tau_i)^{\frac{1}{1-\gamma}}\, di\right)^{\gamma}}.$$

Implication of τ in the Simple Model

- Now, suppose that a_i and $1-\tau_i$ are jointly lognormal, i.e.:

$$(\log(a_i), \log(1-\tau_i)) \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma}), \text{ where:}$$

$$\boldsymbol{\mu} = \left[\nu_a - \sigma_a^2/2, \nu_\tau - \sigma_\tau^2/2\right]; \qquad \boldsymbol{\Sigma} = \begin{bmatrix} \sigma_a^2 & \rho\sigma_a\sigma_\tau \\ \rho\sigma_a\sigma_\tau & \sigma_\tau^2 \end{bmatrix}$$

- Then:

$$A = \exp\left(\nu_a + \frac{\gamma}{1-\gamma}\frac{1}{2}(\sigma^a - \sigma_\tau^2)\right)$$

- If $\sigma_\tau = 0$, back to previous case!
- In this setting, ν_τ does not influence A:
  1. it does not distort the allocation of input across firms
  2. the total supply of inputs is fixed (no investment, no labor supply decision)
- But dispersion in tax rates is bad for aggregate productivity!
- Some productive firms do not expand, because $(1-\tau_i)\, a_i$ is low even when a_i is large.

Quantitative Study of Distortions: Hopenhayn and Rogerson (1993)

- **Q:** What is the effect of firing taxes on employment, output, and productivity?
- Answering this question needs a framework with firing/hiring behavior of firms.
- The employment effect is not obvious.
  - A firing tax makes the firms fire less (employment goes up) but also makes the firm reluctant to hire (employment goes down)
  - It is a quantitative question to see which effect is stronger.
- The paper extends the Hopenhayn (1992) framework to general equilibrium.

Firms

- Firms act competitively in both product and labor market.
- The wage is numéraire. Firm's period profit:

$$pf(n', z) - n' - pc_f - g(n', n)$$

- Firing tax:

$$g(n', n) = \tau \max(0, n - n')$$

- Entry ("free entry"):
  - Many potential entrants.
  - An entrant pays c_e units of goods to enter.
  - Draws z from distribution $\nu(z)$.

Timing: Incumbents and Entrants

Incumbents

- At the beginning of period t, an incumbent firm (operated in period $t-1$) has the state variables (z_{-1}, n).
- It makes the exit decision:
  - If exit, pay $-g(0, n)$.
  - If stay, draw the new z from $F(z_{-1}, z)$.
- Denote the measure of firms at this timing as $\mu(z, n)$.
- Decide n', pay pc_f, and produce.

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Entrants

- At the beginning of period t, pay pc_e and enter.
- Draw s from $\nu(z)$.
- Hire n' and produce.

Firm's Decision Problem

- Firms optimize:

$$W(z,n;p) = \max_{n' \geq 0} \left[pf(n',z) - n' - pc_f - g(n',n) + \beta \max\left(E_z W(z',n';p), -g(0,n')\right) \right]$$

- Decision rule for today's employment:

$$n' = N(z,n;p)$$

- Decision rule for exit in the beginning of next period:

$$X(z,n,p) = \begin{cases} 1, & \text{exit} \\ 0, & \text{stay} \end{cases}$$

- Free entry condition:

$$W^e(p) = pc_e$$

- When there is positive entry:

$$W^e(p) = \int W(z,0,p) d\nu(z)$$

Closing the Model

- Representative consumer:

$$\max \sum_{t=1}^{\infty} \beta^t \left[u\left(c_t\right) - AN_t\right]$$

- Government collects firing taxes and lump-sum transfer back to consumers.
- Given prices, firms and households optimize. There is free entry.
- The distribution of firms is stationary
  - Don't forget to keep track of entry and exit: every period they must coincide
- Markets clear: here only one, total labor supply = total labor demand (+goods market)

Calibration

- Functional forms:
  - Production function: $f(n, z) = zn^{\theta}$.
  - Utility function: $u(c) = \log(c)$.
- Stochastic process:

$$\log(z_t) = a + \rho \log(z_{t-1}) + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \sigma_{\epsilon}^2)$$

- Parameters from outside the model:
  - $\theta = 0.64$ from labor share, $\beta = 0.8$ assuming one period represents 5 years.
- Parameter estimation based on U.S. economy with $\tau = 0$:
  - A: Target employment $N = 0.6$.
  - a: Match the average firm size in the U.S.
  - c_f: Match establishment exit rate in the U.S.
  - c_e: Normalize $p = 1$ so that $W^e(1) = c_e$.

Policy Experiment

$\Longrightarrow$ *Firms face a tax on job destruction modeled as an adjustment cost*

$$g(n_t, n_{t-1}) \;=\; \tau \;\max\{0,\; n_{t-1} - n_t\}$$ (in wage units; calibrated so $\tau \approx$ "one year's wage").

Behavioral effects:

- Firms become more cautious about job creation $\Rightarrow$ fewer employment adjustments overall.
- Employment becomes more persistent, with $\downarrow$ variance of growth rates and $\downarrow$ job turnover.
- The policy creates *inaction bands* (adjust less often, but by more when they do adjust).
- Absent adjustment costs, employment is given by:

$$n' = \eta(z, n; p) = (\alpha\, z\, p)^{\frac{1}{1-\alpha}}, \text{ independent of } n$$

  (log employment proportional to log productivity)
- If adjustment costs are positive, then employment:

Policy Experiment

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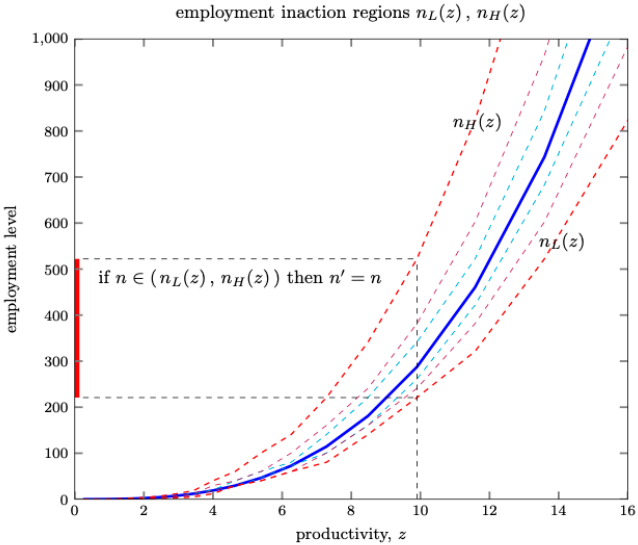
GE effects:

1. A dismissal tax = 1yr’s wage reduces long-run employment by $\approx$ 2.5%.
2. Moving from $\tau = 0$ to the policy level reduces the job destruction rate
3. Average labor productivity $\downarrow$, generating a **welfare loss**.

- *firing cost distorts the reallocation/adjustment margin*

The Inaction Region



Innovation

Klette–Kortum (2004): What changes relative to Hopenhayn?

- You already know Hopenhayn: firm state a (productivity), Markov shocks, entry/exit, stationary distribution, selection & reallocation.
- Klette–Kortum replaces exogenous productivity dynamics with **endogenous innovation** and **creative destruction**.

State variable:

$$\text{Hopenhayn: } a \quad \rightsquigarrow \quad \text{KK: } n \in \{0, 1, 2, \dots\} \text{ (number of product lines)}$$

Interpretation: a firm is a portfolio of product lines; innovation adds lines; competitor innovation destroys lines.

Product lines and reduced-form profits

- Continuum of product lines $j \in [0, 1]$; each line has a current leader (monopolist).
- Reduced form: each *active* line yields a constant per-period profit $\pi > 0$ to its leader.

A firm with n lines earns per-period profits

$$\Pi(n) = n\,\pi.$$

Creative destruction at the line level:

- Let τ be the *per-period probability that a given line is displaced* by someone else's innovation.
- Then a firm with n lines has total expected line losses $n\tau$ per period (first-order in τ).

Innovation technology

Discrete time, discount factor $\beta \in (0, 1)$.

- A firm chooses **per-line** innovation intensity $x \geq 0$ each period.
- Interpretation: in a period, each existing line generates a successful innovation with (small) probability x.
- With n lines: expected innovations (new lines) are nx (first-order).

R&D costs scale with scope:

$$\text{R\&D cost}(n, x) = n\, w\, \psi(x), \qquad \psi' > 0,\ \psi'' > 0.$$

Key device: everything scales with $n \Rightarrow$ tractable firm problem and proportionality of R&D to size.

Bellman equation and linearity in n

Let V_n be the value of a firm with n product lines; $V_0 = 0$ (exit).

Birth–death approximation (first-order in x, τ):

$$\Pr(n \to n+1) = nx, \qquad \Pr(n \to n-1) = n\tau, \qquad \Pr(n \to n) = 1 - nx - n\tau.$$

Bellman equation (for $n \geq 1$):

$$V_n = \max_{x \geq 0} \left\{ n\pi - nw\psi(x) + \beta \Big[(nx)V_{n+1} + (n\tau)V_{n-1} + (1 - nx - n\tau)V_n \Big] \right\}.$$

Guess-and-verify: $V_n = nv$ (linearity). Then $V_{n+1} - V_n = v$ and $V_{n-1} - V_n = -v$ and the Bellman collapses to a scalar equation for v.

Reduced scalar problem and optimal x

Plug $V_n = nv$ into the Bellman and simplify:

$$(1-\beta)\, v = \max_{x \geq 0} \left\{ \pi - w\psi(x) + \beta\, v\,(x - \tau) \right\}.$$

FOC (interior):

$$w\psi'(x) = \beta v.$$

Scaling implications (central KK result):

- x is *independent of* n (innovation intensity per line constant in equilibrium).
- Total innovations and total R&D scale with firm size:

$$\lambda(n) = nx, \qquad \text{R\&D}(n) = n\, w\, \psi(x).$$

Stationary firm-size distribution

Let p_n be the stationary mass of firms of size $n \geq 1$.

With birth and death probabilities

$$\lambda_n \equiv nx, \qquad \mu_n \equiv n\tau,$$

the (reversible) birth–death chain satisfies detailed balance:

$$p_n \, \lambda_n = p_{n+1} \, \mu_{n+1} \quad \Rightarrow \quad p_{n+1} = p_n \, \frac{nx}{(n+1)\tau}.$$

Iterating from p_1 yields

$$p_n = p_1 \, \frac{\rho^{n-1}}{n}, \qquad \rho \equiv \frac{x}{\tau}.$$

Normalization uses

$$\sum_{n \geq 1} \frac{\rho^{n-1}}{n} = \frac{-\ln(1-\rho)}{\rho} \quad (\rho \in (0,1)), \qquad p_1 = \frac{\rho}{-\ln(1-\rho)}.$$

Shape: $p_n \propto \rho^n / n \Rightarrow$ highly skewed size distribution (concentration among large firms).

Equilibrium closure: τ and entry

Creative destruction hazard τ is endogenous.

Because each innovation targets a random existing line, τ is pinned down by aggregate innovation effort per line:

$$\tau = \underbrace{\sum_{n \geq 1} p_n (nx)}_{\text{incumbent innovations per period}} + \underbrace{M_e x_e}_{\text{entrant innovations per period}},$$

where M_e is entrant mass and x_e is entrant intensity (or equivalently an entrant success probability).

Free entry (Hopenhayn analogue): If entry requires fixed cost f_e, then

$$\beta V_1 \leq f_e \quad \text{with equality if } M_e > 0.$$

This pins down v (hence x via $w\psi'(x) = \beta v$) jointly with τ.

Aggregate growth and connection to FHK reallocation

Quality Ladders:

- An innovation displaces the incumbent leader of a line and raises that line's quality by factor $\Lambda > 1$.
- A given line is hit (and upgraded) with probability τ each period.

Then expected log growth of aggregate technology is

$$g_A \equiv \mathbb{E}\left[\ln A_{t+1} - \ln A_t\right] = \tau \ln(\Lambda).$$

Foster–Haltiwanger–Krizan: Each innovation event is simultaneously

- *within-line* productivity growth (quality jump), and
- *between-firm reallocation* (ownership of the line changes).

So KK gives a structural microfoundation for the **reallocation** component in the empirical productivity-growth decomposition.

The Empirical Implications (and Limitations) of Klette-Kortum

Major Strengths:

- clear definition of "the firm", allows analysis of the firm-size distribution.
- can calculate average growth rate of the firm in closed form:
  - average number of product lines next period per line this period is $1 + x - \tau$
  - expected net growth rate of any firm is $x - \tau$

Empirical Limitations:

1. Average growth rates of all incumbent firms is negative!
   - Because entry rate is positive, and $\tau \propto x + M_e$, $\implies x - \tau < 0$

   $\implies$ model firm-size distribution does not feature a Pareto tail!
2. In the data, firms with very advanced products make higher-than-average profits
   - In the model, a firm with a higher-quality product does not earn a higher profit, because:
   - (i) the elasticity of substitution across goods is 1, and (ii) the technology is not cumulative